Business Intelligence (BI)

Business Intelligence (BI) refers to the technologies, tools, and processes used to **collect, analyze, and transform raw business data into meaningful information** to support **better decision-making**.

BI helps organizations understand business performance, identify trends, and make strategic decisions based on data rather than intuition.

Brief History of Business Intelligence

1. **Early Concepts (1950s–1960s)**
   The term Business Intelligence was introduced by **Hans Peter Luhn** in **1958**. He described BI as systems that automatically collect and analyze business information.
2. **Decision Support Systems (1970s–1980s)**
   Organizations started using **Decision Support Systems (DSS)** to help managers analyze data and support decision-making.
3. **Data Warehousing Era (1990s)**
   Development of **data warehouses** enabled storage of large amounts of historical data. Tools like **OLAP** were introduced for faster data analysis.
4. **Modern BI (2000s–Present)**
   BI evolved with **big data, cloud computing, and advanced analytics**. Modern tools such as **Microsoft Power BI**, **Tableau**, and **QlikView** provide dashboards, real-time analysis, and predictive insights.

Business Intelligence Architecture for Retail Industry

Business Intelligence (BI) architecture provides a structured framework for collecting, storing, analyzing, and presenting business data to support decision-making.

```
Data Sources
    ↓
ETL (Extract, Transform, Load)
    ↓
Data Warehouse
    ↓
OLAP Server
    ↓
Analytics Layer
    ↓
Reporting & Dashboards
```

1. Data Sources

Data is collected from multiple sources such as:

- Point of Sale (POS) systems
- Customer databases
- Online shopping platforms
- Inventory management systems
- Social media and external market data

These sources provide raw data about sales, customers, and operations.

2. ETL Process (Extract, Transform, Load)

ETL tools extract data from different sources, transform it into a consistent format, and load it into the data warehouse. This step ensures data cleaning, integration, and consistency.

3. Data Warehouse

A **data warehouse** stores large volumes of historical and integrated data.
It organizes data in a structured form, making it suitable for analysis and reporting.

4. Data Analysis (OLAP)

Online Analytical Processing (OLAP) tools analyze data using multidimensional models.
Retail companies can analyze sales by **product, region, time, and customer category**.

5. Reporting & Visualization

Dashboards, charts, and reports present analyzed data in an easy-to-understand format.
BI tools such as **Microsoft Power BI**, **Tableau**, and **QlikView** help managers monitor performance and trends.

6. Decision Support Systems (DSS)

Decision Support Systems use analyzed data and reports to help managers make **strategic and operational decisions** such as pricing, marketing strategies, and inventory planning.

Justification

- Helps retailers **analyze sales trends and customer behavior**.
- Supports **inventory optimization and demand forecasting**.
- Improves **strategic decision-making and operational efficiency**.
- Provides **real-time insights through dashboards and reports**.

Business Intelligence Scenarios

A **Business Intelligence (BI) scenario** refers to a real-world business situation where BI tools and techniques are used to **analyze data and support decision-making**. Organizations use BI to study large datasets, identify trends, and make **strategic business decisions**.

BI scenarios generally involve:

- Large volumes of **structured and unstructured data**
- **Performance monitoring**
- **Predictive analysis**
- **Strategic planning**

Example: Retail Industry BI Scenario

Business Problem

A retail company observes **declining sales in certain regions** and wants to identify the reasons and improve performance.

BI Tasks Used

1. **Data Collection** – Gather sales data from different regions and time periods.
2. **Data Analysis** – Use BI tools to analyze sales by product, region, and season.
3. **Performance Monitoring** – Identify low-performing products and regions.
4. **Trend Analysis** – Study seasonal patterns and customer buying behavior.
5. **Visualization** – Display results using dashboards and reports.

Mathematical Analysis

To analyze sales performance, the **growth rate** of sales can be calculated.

Growth Rate={Previous Sales - Current Sales}/ Previous Sales×100

If the growth rate is **negative**, it indicates a decline in sales.

Strategic Decision-Making

Using BI insights, the company can:

- Improve **marketing strategies in weak regions**
- Adjust **inventory and product supply**
- Introduce **promotions or discounts**
- Focus on **high-demand products**

Advantages of BI Scenarios

- Improves **real-time business monitoring**
- Helps in **data-driven decision making**
- Enhances **customer understanding**
- Supports **predictive analysis and planning**

Disadvantages

- **Data privacy and security issues**
- Requires **high data processing capability**
- Needs **skilled analysts and BI tools**

Goals of Business Intelligence and Future Trends

1. Goals of Business Intelligence

The main goal of **Business Intelligence (BI)** is to transform raw data into meaningful information that helps organizations make better decisions.

1. **Improved Decision Making**
   BI provides accurate and timely information to support strategic and operational decisions.
2. **Performance Monitoring**
   Organizations can track key performance indicators (KPIs) and measure business performance.
3. **Data Integration**
   BI combines data from multiple sources into a single system for better analysis.
4. **Identifying Trends and Patterns**
   Helps businesses detect market trends, customer behavior, and business opportunities.
5. **Competitive Advantage**
   BI enables organizations to make faster and smarter decisions than competitors.

2. Future Trends of Business Intelligence

1. **Artificial Intelligence and Machine Learning Integration**
   BI tools will increasingly use AI and ML for predictive and automated analysis.
2. **Cloud-based BI**
   Cloud platforms will allow businesses to access BI tools from anywhere with better scalability.
3. **Real-Time Analytics**
   Organizations will analyze data instantly for faster decision-making.
4. **Self-Service BI**
   Users will be able to generate reports and dashboards without technical expertise.
5. **Advanced Data Visualization**
   Interactive dashboards and visual analytics will improve understanding of complex data.

Data, Information, and Knowledge in Business Intelligence

In **Business Intelligence (BI)**, data, information, and knowledge represent different levels of understanding that help organizations make better decisions.

1. Data

Data refers to raw, unprocessed facts and figures collected from various sources.
It has no meaning by itself until it is processed.

Examples:

- Sales numbers
- Customer IDs
- Transaction records
- Product prices

In BI, data is collected from sources such as databases, ERP systems, and web applications.

2. Information

Information is processed and organized data that has meaning and context.
When data is analyzed and structured, it becomes useful for understanding business activities.

Example:

- Monthly sales report
- Customer purchase summary
- Regional sales comparison

Information helps managers understand **what is happening in the business**.

3. Knowledge

Knowledge is the insight or understanding gained from analyzing information and applying experience or expertise. It helps organizations make strategic decisions.

Example:

- Identifying that sales increase during festive seasons
- Recognizing that certain products perform better in specific regions
- Deciding to increase marketing in low-performing areas

Knowledge helps businesses determine **what actions should be taken**.

1. Business Intelligence Tasks

Business Intelligence tasks involve activities used to **collect, analyze, and interpret business data** for decision-making.

1. **Data Collection** – Gathering data from sources such as databases, ERP systems, CRM systems, and web applications.
2. **Data Integration** – Combining data from multiple sources into a unified system.
3. **Data Analysis** – Examining data to identify patterns, trends, and relationships.
4. **Data Mining** – Discovering hidden patterns and useful information from large datasets.
5. **Reporting and Visualization** – Presenting insights using dashboards, charts, and reports.
6. **Performance Monitoring** – Tracking key performance indicators (KPIs) to evaluate business performance.

2. Analysis Formats in Business Intelligence

Analysis formats refer to different ways data is analyzed and presented.

1. **Descriptive Analysis** – Explains what has happened in the past using reports and summaries.
2. **Diagnostic Analysis** – Identifies the reasons behind a particular event or trend.
3. **Predictive Analysis** – Uses historical data to predict future outcomes.
4. **Prescriptive Analysis** – Suggests actions or strategies to achieve desired results.

Decision Support System (DSS)

A **Decision Support System (DSS)** is a computer-based information system that helps managers and decision-makers analyze data, evaluate alternatives, and make effective business decisions.

A DSS combines **data, analytical models, and software tools** to support decision-making in complex business situations.

Example:
A company may use a DSS to analyze sales data and determine the **best pricing strategy or marketing plan**.

Simon's Decision Making Process

The decision-making model was proposed by **Herbert A. Simon**. It explains how managers identify problems and select the best solution in a systematic way. The process consists of four main phases.

Representation: Intelligence Phase → Design Phase → Choice Phase → (Feedback / Choosing) → Intelligence Phase

1. Intelligence Phase
In this phase, the problem or opportunity is identified. Data is collected and analyzed to understand the business situation.

2. Design Phase
In this stage, possible solutions or alternatives are developed. Different strategies are analyzed to solve the identified problem.

3. Choice Phase
The best alternative is selected after comparing different solutions and evaluating their effectiveness.

4. Implementation Phase
The chosen solution is implemented and the results are monitored to ensure the decision achieves the desired outcome.

Advantages

1. **Systematic Approach** – Provides a structured method for solving problems.
2. **Better Decision Quality** – Helps select the best alternative after analysis.
3. **Clear Problem Identification** – Helps managers understand problems properly.
4. **Rational Decision Making** – Encourages logical and data-based decisions.

Decision Support System – Business Intelligence Connection

Business Intelligence (BI) and **Decision Support Systems (DSS)** are closely related technologies that help organizations make better decisions. BI focuses on **collecting, processing, and analyzing business data**, while DSS uses this analyzed information to **assist managers in making effective decisions**.

1. Data Processing
BI collects data from multiple sources and processes it using tools such as data warehouses and analytical models. DSS uses this processed data to support decision-making.

2. Information Generation
BI converts raw data into meaningful information through reports, dashboards, and visualization. DSS uses this information to evaluate different alternatives.

3. Analytical Support
BI performs analysis such as trend analysis and data mining. DSS applies these insights to solve complex business problems.

4. Decision-Making Support
BI provides insights about business performance, while DSS helps managers choose the best strategy or action.

Features / Advantages

- **Improves decision-making** using data-driven insights.
- **Faster analysis of large volumes of data.**
- **Better strategic planning and forecasting.**
- **Enhances organizational efficiency and productivity.**

Impact of Business Intelligence on Decision-Making in Organizations

Business Intelligence (BI) helps organizations improve their decision-making processes by converting large amounts of raw data into meaningful insights.

1. Data-Driven Decisions
BI enables managers to make decisions based on accurate data and analysis rather than intuition or guesswork.

2. Faster Decision Making
BI tools provide real-time reports and dashboards, allowing organizations to quickly analyze information and respond to business changes.

3. Improved Business Performance
By analyzing trends and performance indicators, BI helps organizations identify strengths, weaknesses, and opportunities for improvement.

4. Better Strategic Planning
BI supports long-term planning by providing insights into market trends, customer behavior, and future predictions.

5. Enhanced Operational Efficiency
BI helps organizations optimize processes, reduce costs, and improve productivity through better analysis of business data.

6. Competitive Advantage – Helps organizations respond quickly to market changes and stay ahead of competitors.

7. **Better Customer Understanding** – Analyzes customer behavior and preferences to improve services.

Models and Modelling in BI

In **Business Intelligence**, a **model** is a simplified representation of a real-world business situation that helps analyze data and support decision-making.

Modelling is the process of creating and using these models to understand relationships between variables, predict future outcomes, and evaluate business strategies.

Models help organizations **analyze complex data, simulate business scenarios, and support strategic planning.**

Model building in Business Intelligence refers to the process of creating analytical or mathematical models that help organizations analyze data, understand patterns, and support decision-makin

Key Components of Model Building in BI

1. Problem Definition
Clearly identifying the business problem or objective that the model needs to solve.

2. Data Collection
Gathering relevant data from sources such as databases, data warehouses, and external systems.

3. Data Preparation
Cleaning, transforming, and organizing data to remove errors and ensure consistency.

4. Model Development
Applying statistical, mathematical, or analytical techniques to build the model.

5. Model Evaluation and Implementation
Testing the model for accuracy and using it to support business decision-making.

Model Presentation

Model presentation refers to the process of presenting the results and outputs of a model in a clear and understandable form so that decision-makers can easily interpret the insights.

The purpose is to make complex analytical results **simple and easy to understand**.

Effective model presentation helps managers **interpret data insights and make informed decisions**.

Steps in Model Presentation

1. Prepare Model Results
The results obtained from the model are organized and summarized for better understanding.

2. Select Presentation Format
Appropriate formats such as **charts, graphs, tables, dashboards, or reports** are chosen to present the model outputs.

3. Visual Representation
The results are displayed visually to highlight important patterns, trends, and relationships in the data.

4. Interpretation and Communication
The presented results are explained to decision-makers so they can understand the insights and make informed business decisions.

Model Assessment

Model assessment is the process of evaluating how well a model performs in analyzing data and producing accurate

results. It helps determine whether the model is reliable, effective, and suitable for supporting business decision-making.

In **Business Intelligence**, model assessment ensures that the model provides **accurate predictions, consistent results, and meaningful insights** based on the available data.

Quality of Models in Business Intelligence

Quality of models in Business Intelligence refers to the degree to which a model accurately represents business data and produces reliable, meaningful results for decision-making.

A high-quality BI model should be **accurate, reliable, simple, and capable of handling large and changing datasets**, ensuring effective analysis and better business decisions.

Factors Contributing to the Quality of Models

1. Accuracy

- Accuracy refers to how correctly the model predicts or represents real-world outcomes.
- A high-quality BI model should produce results that closely match actual business data and trends.

2. Data Quality

- The quality of input data directly affects the model's performance.
- Clean, complete, and consistent data ensures the model generates reliable insights.

3. Simplicity

- A good model should be simple and easy to understand.
- Simple models are easier to interpret, implement, and maintain in business environments.

4. Reliability

- Reliability means the model produces consistent results for similar datasets.
- A reliable model increases confidence among managers and decision-makers.

5. Scalability

- Scalability refers to the model's ability to handle large and growing volumes of data.
- This ensures the model remains effective as business data increases.

6. Flexibility

- Flexible models remain useful even when business requirements evolve.

Modelling using Logical Structures in BI

In **Business Intelligence (BI)**, logical structures are used to organize and represent knowledge in a structured way. They help systems understand relationships between data and support effective analysis and decision-making.

1. Ontology

Ontology is a logical structure that defines the **concepts, categories, and relationships** within a specific domain. It provides a common vocabulary for representing knowledge.

In BI modeling, ontology helps:

- Organize business data into meaningful concepts.
- Define relationships between entities such as customers, products, and sales.
- Improve data integration and knowledge sharing.

Example:
In a retail BI system, ontology may define relationships such as **Customer → purchases → Product**.

2. Frame

A **frame** is a data structure used to represent knowledge in the form of **attributes (slots) and values** related to an object or concept.

In BI modeling, frames help:

- Store structured information about business entities.
- Represent real-world objects with their properties.
- Support reasoning and decision-making.

Example:
Frame for a Product:
Product Name, Price, Category, Stock Quantity, Supplier.

Modelling using Graph Structure: Business Process Model and Notation (BPMN)

In **Business Intelligence (BI)**, graph structures are used to represent business processes and relationships visually. One important graph-based modelling technique is **Business Process Model and Notation (BPMN)**.

Business Process Model and Notation (BPMN)

is a **graphical modeling technique** used to represent business processes in the form of flow diagrams. It provides a standard way to visualize how different tasks, events, and decisions are connected in a business workflow.

BPMN helps organizations clearly understand **how business processes operate**, how data flows between activities, and where improvements can be made.

Key Elements of BPMN

1. Events
Events represent something that happens during a process, such as the start or end of a process.

2. Activities (Tasks)
Activities represent the work or tasks performed within a business process.

3. Gateways
Gateways are decision points that control the flow of the process based on certain conditions.

4. Sequence Flow
Sequence flow shows the order in which activities and events occur in a process.

Significance of BPMN in BI Modelling

1. Process Visualization
BPMN provides a visual representation of business processes, making it easier to understand the flow of activities and data.

2. Better Process Analysis
It helps analysts examine business workflows and identify inefficiencies or bottlenecks in processes.

3. Improved Communication
BPMN diagrams provide a common language that can be easily understood by business analysts, managers, and technical teams.

4. Process Optimization
Organizations can analyze BPMN models to improve efficiency, reduce costs, and optimize business operations.

5. Support for Decision Making
By clearly showing how processes work, BPMN helps managers make better operational and strategic decisions.

Using Business Process Model and Notation (BPMN) to Model a Business Process

Business Process Model and Notation (BPMN) is a graphical method used to represent business workflows using standardized symbols such as events, tasks, and gateways. It helps organizations visualize and analyze complex business processes.

Example: Online Order Processing System

Consider an **online retail company** that processes customer orders. BPMN can be used to model the entire process from order placement to delivery.

Steps in the BPMN Model

1. **Start Event** – The process begins when a customer places an order on the website.
2. **Order Processing Task** – The system records the order details and verifies customer information.
3. **Payment Verification (Gateway)** – The system checks whether the payment is successful.
    - If payment fails, the order is cancelled.
    - If payment succeeds, the process continues.
4. **Inventory Check Task** – The system verifies whether the ordered product is available in stock.
5. **Order Packaging and Shipping** – If the product is available, the warehouse packages the item and sends it for delivery.
6. **End Event** – The process ends when the product is successfully delivered to the customer.

Benefits of Using BPMN

- Provides a **clear visual representation** of business processes.
- Helps identify **inefficiencies and bottlenecks** in workflows.
- Improves **communication between business and technical teams**.
- Supports **process improvement and better decision-making**.

1. Modelling using Probabilistic Structures

Probabilistic modelling is a technique used in Business Intelligence where uncertainty and randomness in data are represented using probability. It helps organizations predict possible outcomes based on historical data.

In this approach, models use **probability distributions and statistical methods** to estimate the likelihood of different events occurring. This is useful when business decisions involve uncertainty.

Applications in BI:

- Risk analysis in banking and finance
- Sales forecasting
- Customer behavior prediction
- Fraud detection

Thus, probabilistic models help businesses make **informed decisions by analyzing the probability of different outcomes.**

2. Modelling using Analytical Structures

Analytical modelling in Business Intelligence refers to the use of mathematical and statistical techniques to analyze data and support decision-making.

These models examine relationships between variables and help organizations understand patterns, trends, and performance indicators.

Applications in BI:

- Sales trend analysis
- Market demand forecasting
- Performance measurement
- Strategic planning

Analytical models enable organizations to **interpret large amounts of data and generate insights for better business strategies.**

1. Data Generation (4 Marks)

- **Data generation** refers to the process of creating or collecting data from various sources for Business Intelligence analysis.
- Data can be generated from **transaction systems, databases, ERP systems, sensors, and web applications**.
- It may also come from **external sources** such as social media, market reports, and online platforms.
- The generated data is stored in **data warehouses or databases** for further processing and analysis.

2. Role of Time (4 Marks)

- **Time** is an important factor in Business Intelligence because business data changes continuously over time.
- It helps organizations analyze **historical data and track performance** over different time periods.
- Time-based analysis allows businesses to identify **trends, seasonal patterns, and growth rates**.
- It also supports **forecasting and strategic planning** based on past performance.

3. Data Quality (4 Marks)

- **Data quality** refers to the accuracy, completeness, consistency, and reliability of data used in BI systems.
- High-quality data ensures that the analysis results are **accurate and meaningful**.
- Poor data quality can lead to **incorrect insights and poor business decisions**.
- Organizations improve data quality through **data cleaning, validation, and integration processes**.